

CHAPTER 10



ESTIMATION AND HYPOTHESIS TESTING: TWO POPULATIONS

Opening Example

Are you planning to take a vacation either just to relax, or just to get away from work and school for a few days? Is taking a vacation very important to you? Vacation may not be equally important to all of us. A recent survey by Access America suggests that taking vacations is more important to the younger generation. (See Case Study 10-2).

10.1

Inferences About the Difference Between Two Population Means for Independent Samples: σ_1 and σ_2 Known

- Independent versus Dependent Samples
- Mean, Standard Deviation, and Sampling Distribution of $\bar{x}_1 - \bar{x}_2$
- Interval Estimation of $\mu_1 - \mu_2$
- Hypothesis Testing About $\mu_1 - \mu_2$

Independent versus Dependent Samples

Definition

Two samples drawn from two populations are **independent** if the selection of one sample from one population does not affect the selection of the second sample from the second population. Otherwise, the samples are **dependent**.

Example 10-1

Suppose we want to estimate the difference between the mean salaries of all male and all female executives. To do so, we draw two samples, one from the population of male executives and another from the population of female executives. These two samples are **independent** because they are drawn from two different populations, and the samples have no effect on each other.

Example 10-2

Suppose we want to estimate the difference between the mean weights of all participants before and after a weight loss program. To accomplish this, suppose we take a sample of 40 participants and measure their weights before and after the completion of this program. Note that these two samples include the same 40 participants. This is an example of two **dependent** samples. Such samples are also called **paired** or **matched samples**.

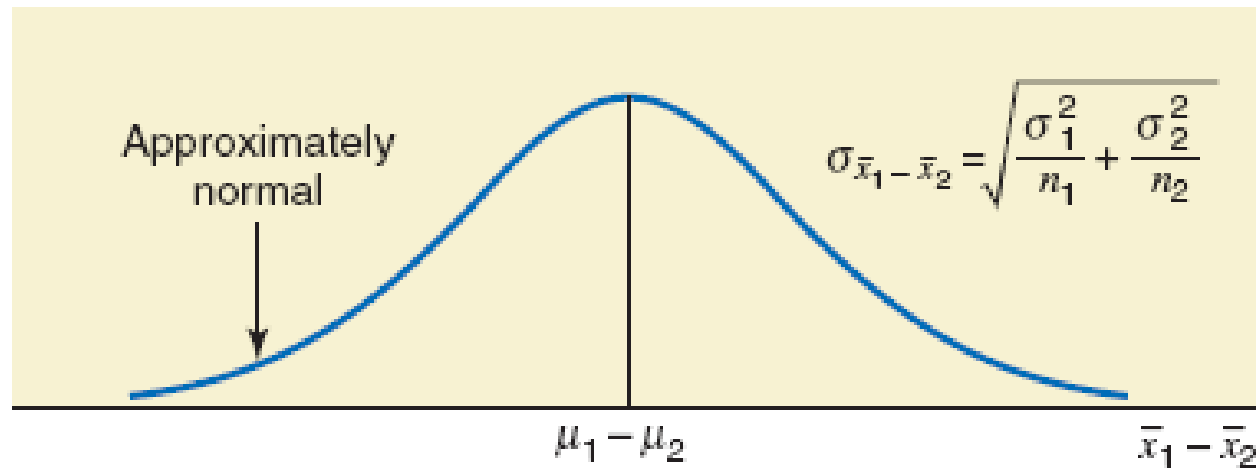
Mean, Standard Deviation, and Sampling Distribution of $\bar{x}_1 - \bar{x}_2$

Three conditions:

1. The two samples are independent
2. The standard deviations σ_1 and σ_2 of the two populations are known
3. At least one of the following conditions is fulfilled:
 - i. Both samples are large (i.e., $n_1 \geq 30$ and $n_2 \geq 30$)
 - ii. If either one or both sample sizes are small, then both populations from which the samples are drawn are normally distributed

Then the sampling distribution of $\bar{x}_1 - \bar{x}_2$ is (approximately) normally distributed.

Figure 10.1



Sampling Distribution, Mean, and Standard Deviation of $\bar{x}_1 - \bar{x}_2$

When the conditions listed on the previous page are satisfied, the **sampling distribution of** $\bar{x}_1 - \bar{x}_2$ is (approximately) normal with its **mean** and **standard deviation** as follows:

$$\mu_{\bar{x}_1 - \bar{x}_2} = \mu_1 - \mu_2 \quad \text{and} \quad \sigma_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

Confidence Interval for $\mu_1 - \mu_2$

When using the normal distribution, the $(1-\alpha)100\%$ **confidence interval for $\mu_1 - \mu_2$** is

$$(\bar{x}_1 - \bar{x}_2) \pm z\sigma_{\bar{x}_1 - \bar{x}_2}$$

The value of z is obtained from the normal distribution table for the given confidence level. The value of $\sigma_{\bar{x}_1 - \bar{x}_2}$ is calculated as explained earlier. Here $\bar{x}_1 - \bar{x}_2$ is the point estimator of $\mu_1 - \mu_2$

Example 10-3

A 2008 survey of low- and middle-income households conducted by Demos, a liberal public group, showed that consumers aged 65 years and older had an average credit card debt of \$10,235 and consumers in the 50- to 64-year age group had an average credit card debt of \$9342 at the time of the survey (*USA TODAY*, July 28, 2009). Suppose that these averages were based on random samples of 1200 and 1400 people for the two groups, respectively. Further assume that the population standard deviations for the two groups were \$2800 and \$2500, respectively.

Example 10-3

Let μ_1 and μ_2 be the respective population means for the two groups, people aged 65 years and older and people in the 50-to 64-year age group.

- a) What is the point estimate of $\mu_1 - \mu_2$?
- b) Construct a 97% confidence interval for $\mu_1 - \mu_2$.

Example 10-3: Solution

(a)

$$\begin{aligned}\text{Point estimate of } \mu_1 - \mu_2 &= \bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2 \\ &= \$10,235 - \$9342 \\ &= \mathbf{\$893}\end{aligned}$$

Example 10-3: Solution

(b)

$$\begin{aligned}\sigma_{\bar{x}_1 - \bar{x}_2} &= \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} = \sqrt{\frac{(2800)^2}{1200} + \frac{(2500)^2}{1400}} \\ &= \mathbf{\$104.8695335}\end{aligned}$$

Example 10-3: Solution

(b)

- 97% confidence level. The value of z is 2.17

$$\begin{aligned}(\bar{x}_1 - \bar{x}_2) \pm z\sigma_{\bar{x}_1 - \bar{x}_2} &= (\$10,235 - \$9342) \pm 2.17(104.8695335) \\ &= 893 \pm 227.57 = \$665.43 \text{ to } \$1120.57\end{aligned}$$

- Thus, with 97% confidence we can state that the difference between the means of 2008 credit card debts for the two groups is between \$665.43 to \$1120.57.

Hypothesis Testing About $\mu_1 - \mu_2$

1. Testing an alternative hypothesis that the means of two populations are different is equivalent to $\mu_1 \neq \mu_2$, which is the same as $\mu_1 - \mu_2 \neq 0$.

HYPOTHESIS TESTING ABOUT $\mu_1 - \mu_2$

2. Testing an alternative hypothesis that the mean of the first population is greater than the mean of the second population is equivalent to $\mu_1 > \mu_2$, which is the same as $\mu_1 - \mu_2 > 0$.

HYPOTHESIS TESTING ABOUT $\mu_1 - \mu_2$

3. Testing an alternative hypothesis that the mean of the first population is less than the mean of the second population is equivalent to $\mu_1 < \mu_2$, which is the same as $\mu_1 - \mu_2 < 0$.

HYPOTHESIS TESTING ABOUT $\mu_1 - \mu_2$

Test Statistic z for $\bar{x}_1 - \bar{x}_2$

When using the normal distribution, the value of the **test statistic z** for $\bar{x}_1 - \bar{x}_2$ is computed as

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sigma_{\bar{x}_1 - \bar{x}_2}}$$

The value of $\mu_1 - \mu_2$ is substituted from H_0 .
The value of $\sigma_{\bar{x}_1 - \bar{x}_2}$ is calculated as earlier in this section.

Example 10-4

Refer to Example 10-3 about the average 2008 credit card debts for consumers of two age groups. Test at the 1% significance level whether the population means for the 2008 credit card debts for the two groups are different.

Example 10-4: Solution

- Step 1:
- $H_0: \mu_1 - \mu_2 = 0$ (The two population means are not different.)
- $H_1: \mu_1 - \mu_2 \neq 0$ (The two population means are different.)

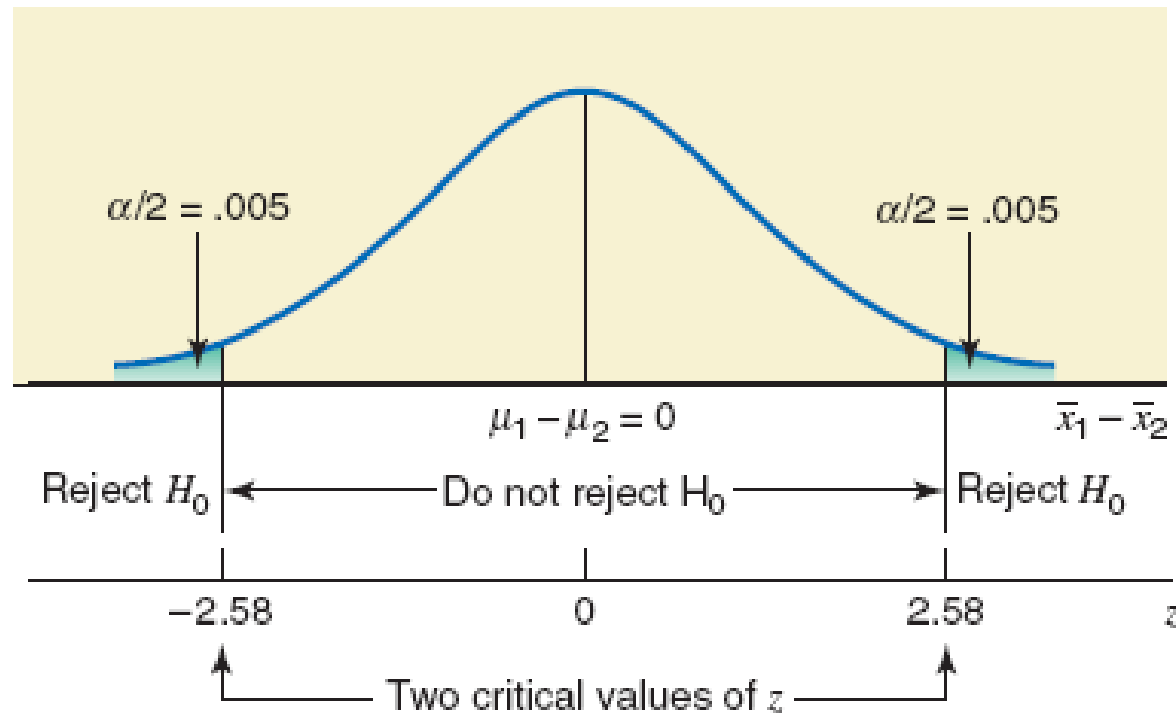
Example 10-4: Solution

- Step 2:
- Population standard deviations, σ_1 and σ_2 are known
- Both samples are large; $n_1 > 30$ and $n_2 > 30$
- Therefore, we use the normal distribution to perform the hypothesis test

Example 10-4: Solution

- Step 3:
- $\alpha = .01$.
- The $\neq$ sign in the alternative hypothesis indicates that the test is two-tailed
- Area in each tail = $\alpha / 2 = .01 / 2 = .005$
- The critical values of z are 2.58 and -2.58

Figure 10.2



Example 10-4: Solution

Step 4:

$$\sigma_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} = \sqrt{\frac{(2800)^2}{1200} + \frac{(2500)^2}{1400}} = \$104.8695335$$

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sigma_{\bar{x}_1 - \bar{x}_2}} = \frac{(10,235 - 9342) - 0}{104.8695335} = 8.52$$

From H_0



Example 10-4: Solution

- Step 5:
- Because the value of the test statistic $z = 8.52$ falls in the rejection region, we reject the null hypothesis H_0 .
- Therefore, we conclude that the mean 1008 credit card debts for the two age groups mentioned in this example are different.

10.2

Inferences About the Difference Between Two Population Means for Independent Samples: σ_1 and σ_2 Unknown but Equal

- Interval Estimation of $\mu_1 - \mu_2$
- Hypothesis Testing About $\mu_1 - \mu_2$

Pooled Standard Deviation for Two Samples

The **pooled standard deviation for two samples** is computed as

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

where n_1 and n_2 are the sizes of the two samples and s_1^2 and s_2^2 are the variances of the two samples, respectively. Here s_p is an estimator of σ .

Estimator of the Standard Deviation of $\bar{x}_1 - \bar{x}_2$

Estimator of the Standard Deviation of $\bar{x}_1 - \bar{x}_2$

The **estimator of the standard deviation**
of $\bar{x}_1 - \bar{x}_2$ is

$$s_{\bar{x}_1 - \bar{x}_2} = s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

Interval Estimation of $\mu_1 - \mu_2$

Confidence Interval for $\mu_1 - \mu_2$

The $(1 - \alpha)100\%$ **confidence interval** for $\mu_1 - \mu_2$ is

$$(\bar{x}_1 - \bar{x}_2) \pm ts_{\bar{x}_1 - \bar{x}_2}$$

where the value of t is obtained from the t distribution table for the given confidence level and $n_1 + n_2 - 2$ degrees of freedom, and $s_{\bar{x}_1 - \bar{x}_2}$ is calculated as explained earlier.

Example 10-5

A consumer agency wanted to estimate the difference in the mean amounts of caffeine in two brands of coffee. The agency took a sample of 15 one-pound jars of Brand I coffee that showed the mean amount of caffeine in these jars to be 80 milligrams per jar with a standard deviation of 5 milligrams. Another sample of 12 one-pound jars of Brand II coffee gave a mean amount of caffeine equal to 77 milligram per jar with a standard deviation of 6 milligrams.

Example 10-5

Construct a 95% confidence interval for the difference between the mean amounts of caffeine in one-pound jars of these two brands of coffee. Assume that the two populations are normally distributed and that the standard deviations of the two populations are equal.

Example 10-5: Solution

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} = \sqrt{\frac{(15 - 1)(5)^2 + (12 - 1)(6)^2}{15 + 12 - 2}} \\ = 5.46260011$$

$$s_{\bar{x}_1 - \bar{x}_2} = s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} = (5.46260011) \sqrt{\frac{1}{15} + \frac{1}{12}} = 2.11565593$$

Example 10-5: Solution

$$\text{Area in each tail} = \alpha/2 = .5 - (.95/2) = .025$$

$$\text{Degrees of freedom} = n_1 + n_2 - 2 = 25$$

$$t = 2.060$$

$$\begin{aligned}(\bar{x}_1 - \bar{x}_2) \pm ts_{\bar{x}_1 - \bar{x}_2} &= (80 - 77) \pm 2.060(2.11565593) \\&= 3 \pm 4.36 \\&= -1.36 \text{ to } 7.36 \text{ milligrams}\end{aligned}$$

Hypothesis Testing About $\mu_1 - \mu_2$

Test Statistic t for $\bar{x}_1 - \bar{x}_2$

The value of the **test statistic t for $\bar{x}_1 - \bar{x}_2$** is computed as

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{S_{\bar{x}_1 - \bar{x}_2}}$$

The value of $\mu_1 - \mu_2$ in this formula is substituted from the null hypothesis and $S_{\bar{x}_1 - \bar{x}_2}$ is calculated as explained earlier.

Example 10-6

A sample of 14 cans of Brand I diet soda gave the mean number of calories of 23 per can with a standard deviation of 3 calories. Another sample of 16 cans of Brand II diet soda gave the mean number of calories of 25 per can with a standard deviation of 4 calories.

Example 10-6

At the 1% significance level, can you conclude that the mean number of calories per can are different for these two brands of diet soda? Assume that the calories per can of diet soda are normally distributed for each of the two brands and that the standard deviations for the two populations are equal.

Example 10-6: Solution

- Step 1:
- $H_0: \mu_1 - \mu_2 = 0$ (The mean numbers of calories are not different.)
- $H_1: \mu_1 - \mu_2 \neq 0$ (The mean numbers of calories are different.)

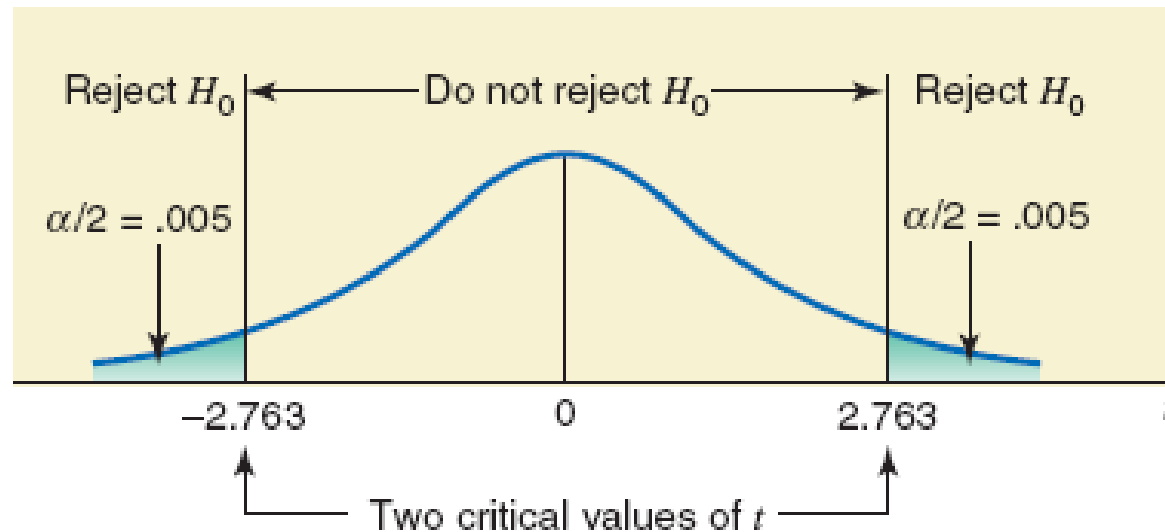
Example 10-6: Solution

- Step 2:
- The two samples are independent
- σ_1 and σ_2 are unknown but equal
- The sample sizes are small but both populations are normally distributed
- We will use the t distribution

Example 10-6: Solution

- Step 3:
- The $\neq$ sign in the alternative hypothesis indicates that the test is two-tailed.
- $\alpha = .01$.
- Area in each tail = $\alpha / 2 = .01 / 2 = .005$
- $df = n_1 + n_2 - 2 = 14 + 16 - 2 = 28$
- Critical values of t are -2.763 and 2.763.

Figure 10.3



Example 10-6: Solution

Step 4:

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} = \sqrt{\frac{(14 - 1)(3)^2 + (16 - 1)(4)^2}{16 + 16 - 2}} = 3.57071421$$

$$s_{\bar{x}_1 - \bar{x}_2} = s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} = (3.57071421) \sqrt{\frac{1}{14} + \frac{1}{16}} = 1.30674760$$

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{s_{\bar{x}_1 - \bar{x}_2}} = \frac{(23 - 25) - 0}{1.30674760} = -1.531$$

Example 10-6: Solution

- Step 5:
- The value of the test statistic $t = -1.531$
 - It falls in the nonrejection region
- Therefore, we fail to reject the null hypothesis
- Consequently, we conclude that there is no difference in the mean numbers of calories per can for the two brands of diet soda.

Example 10-7

A sample of 40 children from New York State showed that the mean time they spend watching television is 28.50 hours per week with a standard deviation of 4 hours. Another sample of 35 children from California showed that the mean time spent by them watching television is 23.25 hours per week with a standard deviation of 5 hours.

Example 10-7

Using a 2.5% significance level, can you conclude that the mean time spent watching television by children in New York State is greater than that for children in California? Assume that the standard deviations for the two populations are equal.

Example 10-7: Solution

- Step 1:
- $H_0: \mu_1 - \mu_2 = 0$
- $H_1: \mu_1 - \mu_2 > 0$

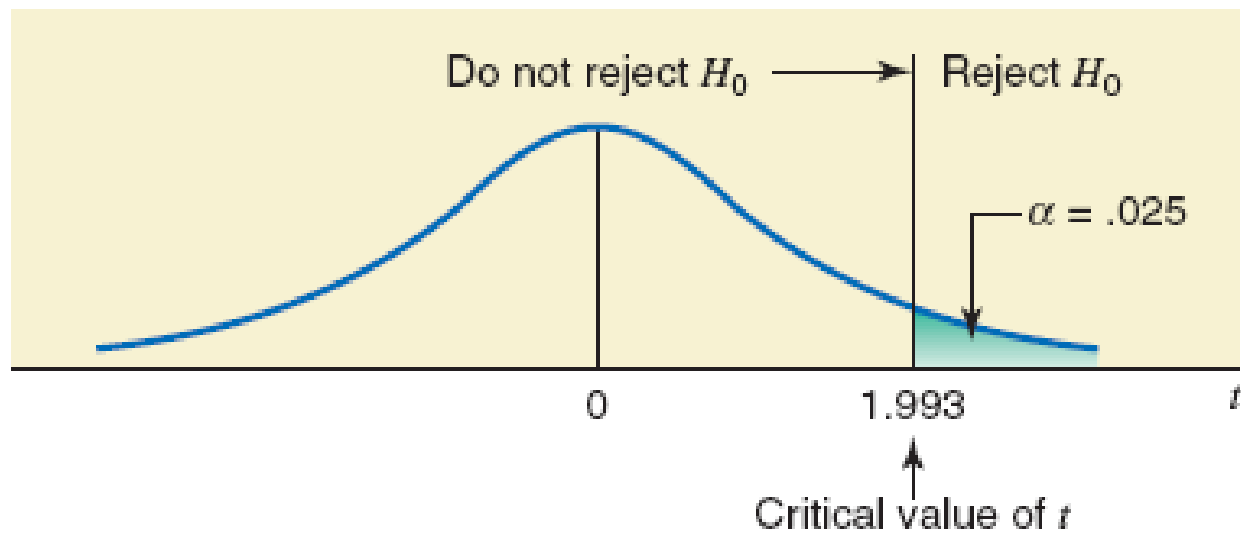
Example 10-7: Solution

- Step 2:
- The two samples are independent
- Standard deviations of the two populations are unknown but assumed to be equal
- Both samples are large
- We use the t distribution to make the test

Example 10-7: Solution

- Step 3:
- $\alpha = .025$
- Area in the right tail of the t distribution = $\alpha = .025$
- $df = n_1 + n_2 - 2 = 40 + 35 - 2 = 73$
- Critical value of t is 1.993

Figure 10.4



Example 10-7: Solution

Step 4:

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} = \sqrt{\frac{(40 - 1)(4)^2 + (35 - 1)(5)^2}{40 + 35 - 2}} = 4.49352655$$

$$s_{\bar{x}_1 - \bar{x}_2} = s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} = (4.49352655) \sqrt{\frac{1}{40} + \frac{1}{35}} = 1.04004930$$

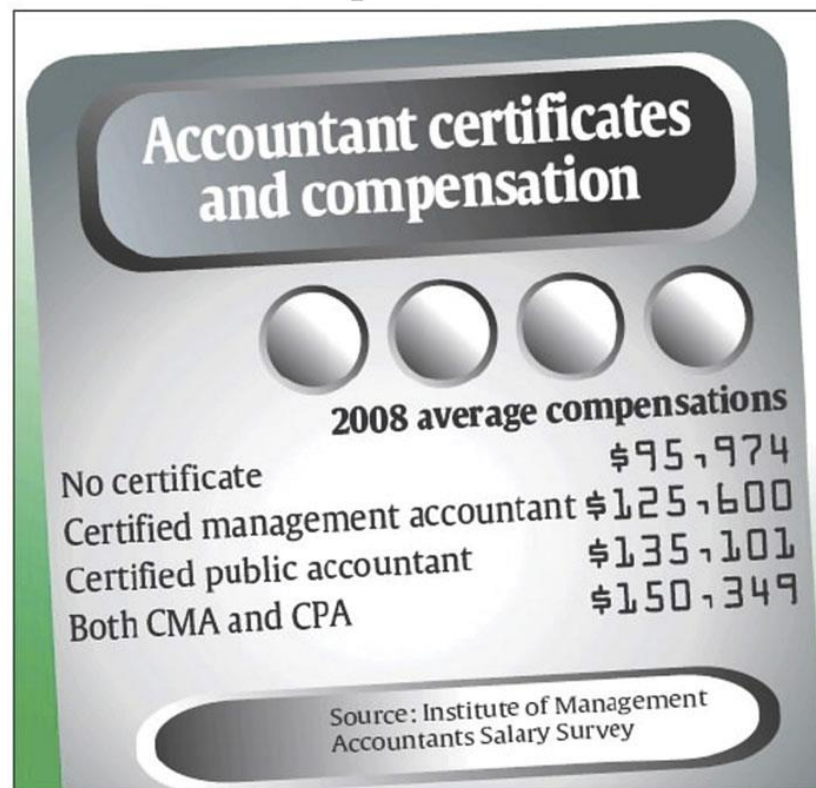
$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{s_{\bar{x}_1 - \bar{x}_2}} = \frac{(28.50 - 23.25) - 0}{1.04004930} = 5.048$$

Example 10-7: Solution

- Step 5:
- The value of the test statistic $t = 5.048$
 - It falls in the rejection region
- Therefore, we reject the null hypothesis H_0
- Hence, we conclude that children in New York State spend more time, on average, watching TV than children in California.

Case Study 10-1 Average Compensation For Accountants

USA TODAY Snapshots®



By Jae Yang and Julie Snider, USA TODAY

Stop here 26.05.2025

10.3

Inferences About the Difference Between Two Population Means for Independent Samples: σ_1 and σ_2 Unknown and Unequal

- Interval Estimation of $\mu_1 - \mu_2$
- Hypothesis Testing About $\mu_1 - \mu_2$

INFERENCES ABOUT THE DIFFERENCE BETWEEN TWO POPULATION MEANS FOR INDEPENDENT SAMPLES: σ_1 AND σ_2 UNKNOWN AND UNEQUAL

Degrees of Freedom If

- The two samples are independent
- The standard deviations σ_1 and σ_2 of are unknown and unequal
- At least one of the following two conditions is fulfilled:
 - i. Both sample are large (i.e., $n_1 \geq 30$ and $n_2 \geq 30$)
 - ii. If either one or both sample sizes are small, then both populations from which the samples are drawn are normally distributed

INFERENCES ABOUT THE DIFFERENCE BETWEEN TWO POPULATION MEANS FOR INDEPENDENT SAMPLES: σ_1 AND σ_2 UNKNOWN AND UNEQUAL

Degrees of Freedom

then the t distribution is used to make inferences about $\mu_1 - \mu_2$ and the **degrees of freedom** for the t distribution are given by

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2} \right)^2}{\frac{\left(\frac{s_1^2}{n_1} \right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2} \right)^2}{n_2 - 1}}$$

The number given by this formula is always rounded down for df .

INFERENCES ABOUT THE DIFFERENCE BETWEEN TWO
POPULATION MEANS FOR INDEPENDENT SAMPLES: σ_1 AND σ_2
UNKNOWN AND UNEQUAL

Estimate of the Standard Deviation of $\bar{x}_1 - \bar{x}_2$

The value of $\mathbf{s}_{\bar{x}_1 - \bar{x}_2}$, is calculated as

$$\mathbf{s}_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{\mathbf{s}_1^2}{n_1} + \frac{\mathbf{s}_2^2}{n_2}}$$

Confidence Interval for $\mu_1 - \mu_2$

The $(1 - \alpha)100\%$ **confidence interval** for $\mu_1 - \mu_2$ is

$$(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2) \pm \mathbf{ts}_{\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2}$$

Where the value of t is obtained from the t distribution table for a given confidence level and the degrees of freedom are given by the formula mentioned earlier.

Example 10-8

According to Example 10-5 of Section 10.2.1, a sample of 15 one-pound jars of coffee of Brand I showed that the mean amount of caffeine in these jars is 80 milligrams per jar with a standard deviation of 5 milligrams. Another sample of 12 one-pound coffee jars of Brand II gave a mean amount of caffeine equal to 77 milligrams per jar with a standard deviation of 6 milligrams.

Example 10-8

Construct a 95% confidence interval for the difference between the mean amounts of caffeine in one-pound coffee jars of these two brands. Assume that the two populations are normally distributed and that the standard deviations of the two populations are not equal.

Example 10-8: Solution

$$s_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = \sqrt{\frac{(5)^2}{15} + \frac{(6)^2}{12}} = 2.16024690$$

$$\text{Area in each tail} = \alpha/2 = .025$$

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2 - 1}} = \frac{\left(\frac{(5)^2}{15} + \frac{(6)^2}{12}\right)^2}{\frac{\left(\frac{(5)^2}{15}\right)^2}{15 - 1} + \frac{\left(\frac{(6)^2}{12}\right)^2}{12 - 1}} = 21.42 \approx 21$$

Example 10-8: Solution

$$**t = 2.080**$$

$$\begin{aligned}(\bar{x}_1 - \bar{x}_2) \pm ts_{\bar{x}_1 - \bar{x}_2} &= (80 - 77) \pm 2.080(2.16024690) \\ &= 3 \pm 4.49 = -1.49 \text{ to } 7.49\end{aligned}$$

Hypothesis Testing About $\mu_1 - \mu_2$

Test Statistic t for $\bar{x}_1 - \bar{x}_2$

The value of the **test statistic t for $\bar{x}_1 - \bar{x}_2$** is computed as

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{S_{\bar{x}_1 - \bar{x}_2}}$$

The value of $\mu_1 - \mu_2$ in this formula is substituted from the null hypothesis and $S_{\bar{x}_1 - \bar{x}_2}$ is calculated as explained earlier.

Example 10-9

According to Example 10-6 of Section 10.2.2, a sample of 14 cans of Brand I diet soda gave the mean number of calories per can of 23 with a standard deviation of 3 calories. Another sample of 16 cans of Brand II diet soda gave the mean number of calories as 25 per can with a standard deviation of 4 calories.

Example 10-9

Test at the 1% significance level whether the mean numbers of calories per can of diet soda are different for these two brands. Assume that the calories per can of diet soda are normally distributed for each of these two brands and that the standard deviations for the two populations are not equal.

Example 10-9: Solution

- Step 1:
- $H_0: \mu_1 - \mu_2 = 0$
 - The mean numbers of calories are not different
- $H_1: \mu_1 - \mu_2 \neq 0$
 - The mean numbers of calories are different

Example 10-9: Solution

- Step 2:
- The two samples are independent
- Standard deviations of the two populations are unknown and unequal
- Both populations are normally distributed
- We use the t distribution to make the test

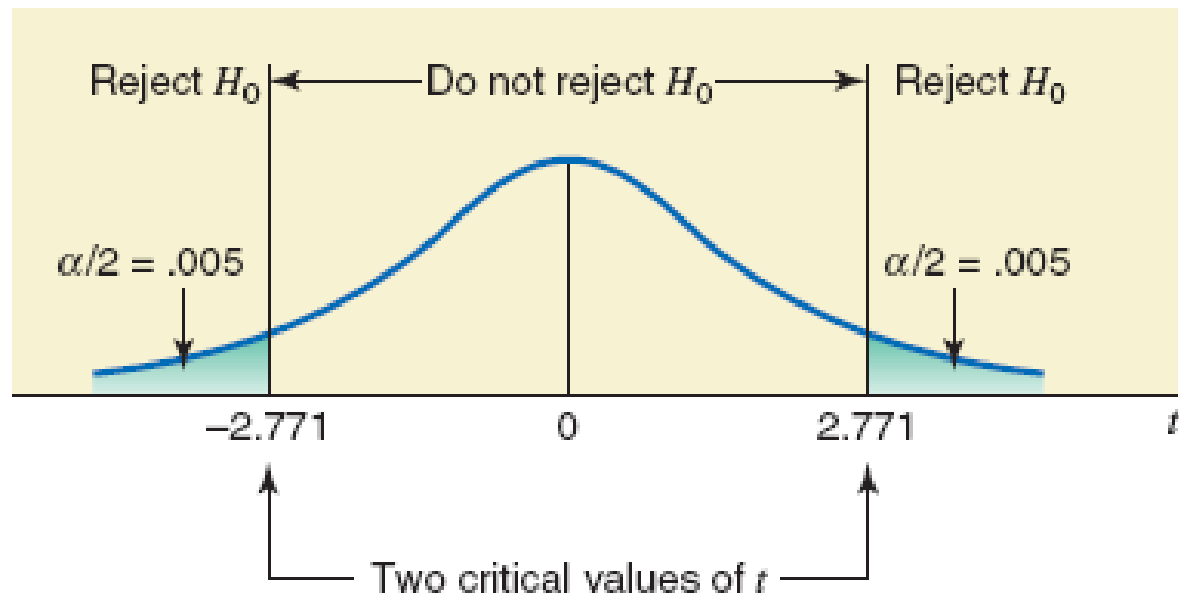
Example 10-9: Solution

- Step 3:
- The $\neq$ in the alternative hypothesis indicates that the test is two-tailed.
- $\alpha = .01$
- Area in each tail = $\alpha / 2 = .01 / 2 = .005$
- The critical values of t are -2.771 and 2.771

Example 10-9: Solution

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2} \right)^2}{\frac{\left(\frac{s_1^2}{n_1} \right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2} \right)^2}{n_2 - 1}} = \frac{\left(\frac{(3)^2}{14} + \frac{(4)^2}{16} \right)^2}{\frac{\left(\frac{(3)^2}{15} \right)^2}{14 - 1} + \frac{\left(\frac{(4)^2}{12} \right)^2}{16 - 1}} = 27.41 \approx 27$$

Figure 10.5



Example 10-9: Solution

Step 4:

$$s_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = \sqrt{\frac{(3)^2}{14} + \frac{(14)^2}{16}} = 1.28173989$$

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{s_{\bar{x}_1 - \bar{x}_2}} = \frac{(23 - 25) - 0}{1.28173989} = -1.560$$

Example 10-9: Solution

- Step 5:
- The test statistic $t = -1.560$
 - It falls in the nonrejection region
- Therefore, we fail to reject the null hypothesis
- Hence, there is no difference in the mean numbers of calories per can for the two brands of diet soda.

10.4

Inferences About the Difference Between Two Population Means for Paired Samples

- Interval Estimation of μ_d
- Hypothesis Testing About μ_d

INFERENCES ABOUT THE DIFFERENCE BETWEEN TWO POPULATION MEANS FOR PAIRED SAMPLES

Definition

Two samples are said to be **paired** or **matched samples** when for each data value collected from one sample there is a corresponding data value collected from the second sample, and both these data values are collected from the same source.

INFERENCES ABOUT THE DIFFERENCE BETWEEN TWO POPULATION MEANS FOR PAIRED SAMPLES

Mean and Standard Deviation of the Paired Differences for Two Samples

The values of the mean and standard deviation, $\bar{d}$ and s_d , respectively, of paired differences for two samples are calculated as

$$\bar{d} = \frac{\sum d}{n}$$
$$s_d = \sqrt{\frac{\sum d^2 - \frac{(\sum d)^2}{n}}{n - 1}}$$

INFERENCES ABOUT THE DIFFERENCE BETWEEN TWO POPULATION MEANS FOR PAIRED SAMPLES

Sampling Distribution, Mean, and Standard Deviation of $\bar{d}$

If σ_d is known and either the sample size is large ($n \geq 30$) or the population is normally distributed, then the **sampling distribution of** $\bar{d}$ is approximately normal with its **mean** and **standard deviation** given as, respectively.

$$\mu_{\bar{d}} = \mu_d \quad \text{and} \quad \sigma_{\bar{d}} = \frac{\sigma_d}{\sqrt{n}}$$

INFERENCES ABOUT THE DIFFERENCE BETWEEN TWO POPULATION MEANS FOR PAIRED SAMPLES

Estimate of the Standard Deviation of Paired Differences If

- The standard deviation σ_d of the population paired differences is unknown
- At least one of the following two conditions is fulfilled:
 - i. The sample size is large (i.e., $n \geq 30$)
 - ii. If the sample size is small, then the population of paired differences is normally distributed

INFERENCES ABOUT THE DIFFERENCE BETWEEN TWO POPULATION MEANS FOR PAIRED SAMPLES

Estimate of the Standard Deviation of Paired Differences

then the t distribution is used to make inferences about μ_d . The standard deviation of $\sigma_{\bar{d}}$ of $\bar{d}$ is estimated by $S_{\bar{d}}$, which is calculated as

$$S_{\bar{d}} = \frac{S_d}{\sqrt{n}}$$

Interval Estimation of μ_d

Confidence Interval for μ_d

The $(1 - \alpha)100\%$ **confidence interval** for μ_d is

$$\bar{d} \pm ts_{\bar{d}}$$

where the value of t is obtained from the t distribution table for the given confidence level and $n - 1$ degrees of freedom, and $s_{\bar{d}}$ is calculated as explained earlier.

Example 10-10

A researcher wanted to find the effect of a special diet on systolic blood pressure. She selected a sample of seven adults and put them on this dietary plan for 3 months. The following table gives the systolic blood pressures (in mm Hg) of these seven adults before and after the completion of this plan.

Example 10-10

Before	210	180	195	220	231	199	224
After	193	186	186	223	220	183	233

Let μ_d be the mean reduction in the systolic blood pressure due to this special dietary plan for the population of all adults. Construct a 95% confidence interval for μ_d . Assume that the population of paired differences is (approximately) normally distributed.

Example 10-10: Solution

Table 10.1

Before	After	Difference	
		d	d^2
210	193	17	289
180	186	-6	36
195	186	9	81
220	223	-3	9
231	220	11	121
199	183	16	256
224	233	-9	81
		$\Sigma d = 35$	$\Sigma d^2 = 873$

Example 10-10: Solution

$$\bar{d} = \frac{\sum d}{n} = \frac{35}{7} = 5.00$$

$$s_d = \sqrt{\frac{\sum d^2 - \frac{(\sum d)^2}{n}}{n-1}} = \sqrt{\frac{873 - \frac{(35)^2}{7}}{7-1}} = 10.78579312$$

Example 10-10: Solution

Hence,

$$s_{\bar{d}} = \frac{s_d}{\sqrt{n}} = \frac{10.78579312}{\sqrt{7}} = 4.07664661$$

$$\text{Area in each tail} = \alpha/2 = .5 - (.95/2) = .025$$

$$df = n - 1 = 7 - 1 = 6$$

$$t = 2.447$$

$$\begin{aligned}\bar{d} \pm ts_{\bar{d}} &= 5.00 \pm 2.447(4.07664661) = 5.00 \pm 9.98 \\ &= -4.98 \text{ to } 14.98\end{aligned}$$

Hypothesis Testing About μ_d

Test Statistic t for $\bar{d}$

The value of the test statistic t for $\bar{d}$ is computed as follows:

$$t = \frac{\bar{d} - \mu_d}{s_{\bar{d}}}$$

Example 10-11

A company wanted to know if attending a course on “how to be a successful salesperson” can increase the average sales of its employees. The company sent six of its salespersons to attend this course. The following table gives the 1-week sales of these salespersons before and after they attended this course.

Example 10-11

Before	12	18	25	9	14	16
After	18	24	24	14	19	20

Using the 1% significance level, can you conclude that the mean weekly sales for all salespersons increase as a result of attending this course? Assume that the population of paired differences has a normal distribution.

Example 10-11: Solution

Table 10.2

Before	After	Difference	
		d	d^2
12	18	-6	36
18	24	-6	36
25	24	1	1
9	14	-5	25
14	19	-5	25
16	20	-4	16
		$\Sigma d = -25$	$\Sigma d^2 = 139$

Example 10-11: Solution

$$\bar{d} = \frac{\sum d}{n} = \frac{-25}{6} = -4.17$$

$$s_d = \sqrt{\frac{\sum d^2 - \frac{(\sum d)^2}{n}}{n-1}} = \sqrt{\frac{139 - \frac{(-25)^2}{6}}{6-1}} = 2.63944439$$

$$s_{\bar{d}} = \frac{s_d}{\sqrt{n}} = \frac{2.63944439}{\sqrt{6}} = 1.07754866$$

Example 10-11: Solution

□ Step 1:

□ $H_0: \mu_d = 0$

■ $\mu_1 - \mu_2 = 0$ or the mean weekly sales do not increase

□ $H_1: \mu_d < 0$

■ $\mu_1 - \mu_2 < 0$ or the mean weekly sales do increase

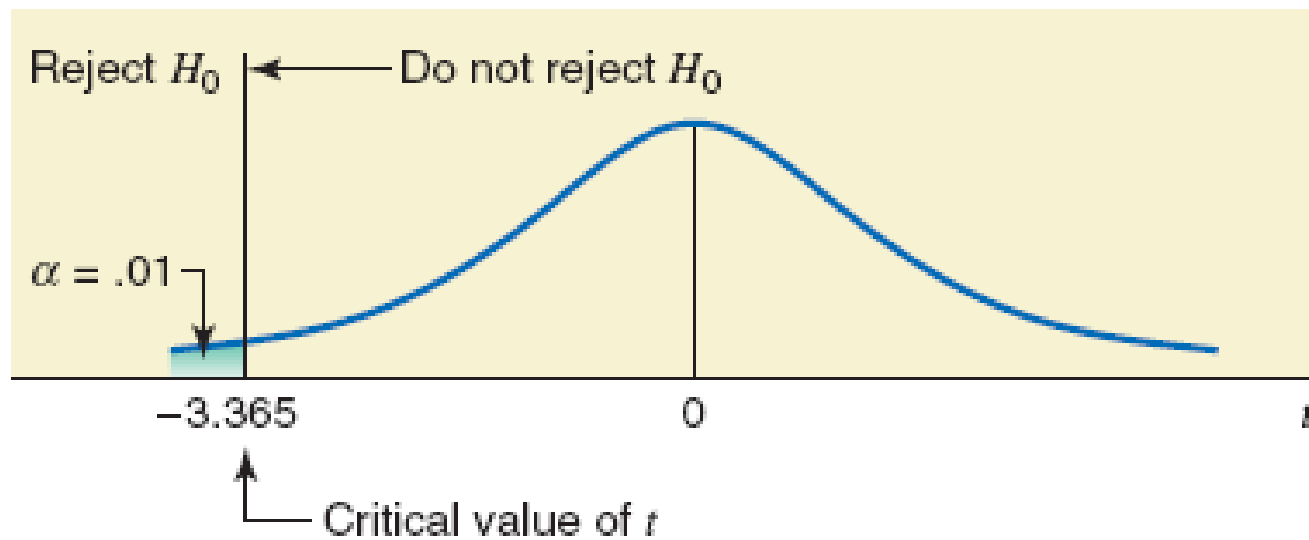
Example 10-11: Solution

- Step 2:
- σ_d is unknown
- The sample size is small ($n < 30$)
- The population of paired differences is normal
- Therefore, we use the t distribution to conduct the test

Example 10-11: Solution

- Step 3:
- $\alpha = .01$.
- Area in left tail = $\alpha = .01$
- $df = n - 1 = 6 - 1 = 5$
- The critical value of t is -3.365

Figure 10.6



Example 10-11: Solution

Step 4:

$$t = \frac{\bar{d} - \mu_d}{s_{\bar{d}}} = \frac{-4.17 - 0}{1.07754866} = -3.870$$

Example 10-11: Solution

- Step 5:
- The value of the test statistic $t = -3.870$
 - It falls in the rejection region
- Therefore, we reject the null hypothesis
- Consequently, we conclude that the mean weekly sales for all salespersons increase as a result of this course.

Example 10-12

Refer to Example 10-10. The table that gives the blood pressures (in mm Hg) of seven adults before and after the completion of a special dietary plan is reproduced here.

Before	210	180	195	220	231	199	224
After	193	186	186	223	220	183	233

Example 10-12

Let μ_d be the mean of the differences between the systolic blood pressures before and after completing this special dietary plan for the population of all adults. Using the 5% significance level, can we conclude that the mean of the paired differences μ_d is difference from zero? Assume that the population of paired differences is (approximately) normally distributed.

Example 10-12: Solution

Table 10-3

Before	After	Difference	
		d	d^2
210	193	17	289
180	186	-6	36
195	186	9	81
220	223	-3	9
231	220	11	121
199	183	16	256
224	233	-9	81
		$\Sigma d = 35$	$\Sigma d^2 = 873$

Example 10-12: Solution

$$\bar{d} = \frac{\sum d}{n} = \frac{35}{7} = 5.00$$

$$s_d = \sqrt{\frac{\sum d^2 - \frac{(\sum d)^2}{n}}{n-1}} = \sqrt{\frac{873 - \frac{(35)^2}{7}}{7-1}} = 10.78579312$$

$$s_{\bar{d}} = \frac{s_d}{\sqrt{n}} = \frac{10.78579312}{\sqrt{7}} = 4.07664661$$

Example 10-12: Solution

□ Step 1:

□ $H_0: \mu_d = 0$

- The mean of the paired differences is not different from zero

□ $H_1: \mu_d \neq 0$

- The mean of the paired differences is different from zero

Example 10-12: Solution

- Step 2:
- σ_d is unknown
- The sample size is small
- The population of paired differences is (approximately) normal
- We use the t distribution to make the test.

Example 10-12: Solution

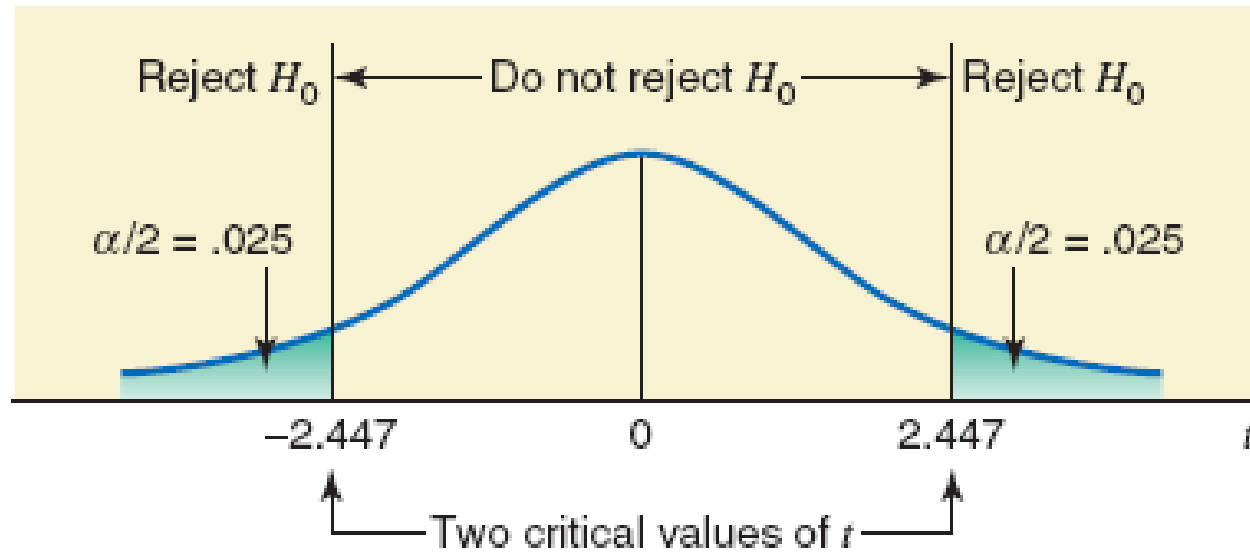
- Step 3:
- $\alpha = .05$
- Area in each tail = $\alpha / 2 = .05 / 2 = .025$
- $df = n - 1 = 7 - 1 = 6$
- The two critical values of t are -2.447 and 2.447

Example 10-12: Solution

Step 4:

$$t = \frac{\bar{d} - \mu_d}{s_{\bar{d}}} = \frac{5.00 - 0}{4.07664661} = 1.226$$

Figure 10.7



Example 10-12: Solution

- Step 5:
- The value of the test statistic $t = 1.226$
- It falls in the nonrejection region
- Therefore, we fail to reject the null hypothesis
- Hence, we conclude that the mean of the population paired difference is not different from zero.

10.5

Inferences About the Difference Between Two Population Proportions for Large and Independent Samples

- Mean, Standard Deviation, and Sampling Distribution of $\hat{p}_1 - \hat{p}_2$
- Interval Estimation of $p_1 - p_2$
- Hypothesis Testing About $p_1 - p_2$

Mean, Standard Deviation, and Sampling Distribution of $\hat{p}_1 - \hat{p}_2$

For two large and independent samples, the **sampling distribution** of $\hat{p}_1 - \hat{p}_2$ is (approximately) normal with its **mean** and **standard deviation** given as

$$\mu_{\hat{p}_1 - \hat{p}_2} = p_1 - p_2$$

and

$$\sigma_{\hat{p}_1 - \hat{p}_2} = \sqrt{\frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}}$$

respectively, where $q_1 = 1 - p_1$ and $q_2 = 1 - p_2$.

Interval Estimation of $p_1 - p_2$

The $(1 - \alpha)100\%$ **confidence interval** for $p_1 - p_2$ is $(\hat{p}_1 - \hat{p}_2) \pm z s_{\hat{p}_1 - \hat{p}_2}$

where the value of z is read from the normal distribution table for the given confidence level, and $s_{\hat{p}_1 - \hat{p}_2}$ is calculated as

$$s_{\hat{p}_1 - \hat{p}_2} = \sqrt{\frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}}$$

Example 10-13

A researcher wanted to estimate the difference between the percentages of users of two toothpastes who will never switch to another toothpaste. In a sample of 500 users of Toothpaste A taken by this researcher, 100 said that they will never switch to another toothpaste. In another sample of 400 users of Toothpaste B taken by the same researcher, 68 said that they will never switch to another toothpaste.

Example 10-13

- a) Let p_1 and p_2 be the proportions of all users of Toothpastes A and B, respectively, who will never switch to another toothpaste. What is the point estimate of $p_1 - p_2$?
- b) Construct a 97% confidence interval for the difference between the proportions of all users of the two toothpastes who will never switch.

Example 10-13: Solution

The two sample proportions are calculated as follows:

$$\hat{p}_1 = x_1 / n_1 = 100 / 500 = .20$$

$$\hat{p}_2 = x_2 / n_2 = 68 / 400 = .17$$

Then,

$$\hat{q}_1 = 1 - .20 = .80 \quad \text{and} \quad \hat{q}_2 = 1 - .17 = .83$$

Example 10-13: Solution

a) Point estimate of $p_1 - p_2$

$$\begin{aligned} &= \hat{p}_1 - \hat{p}_2 \\ &= .20 - .17 = .03 \end{aligned}$$

Example 10-13: Solution

b)

$$n_1 \hat{p}_1 = 500(.20) = 100$$

$$n_1 \hat{q}_1 = 500(.80) = 400$$

$$n_2 \hat{p}_2 = 400(.17) = 68$$

$$n_2 \hat{q}_2 = 400(.83) = 332$$

Example 10-13: Solution

b)

- Each of these values is greater than 5
- Both sample sizes are large
- Consequently, we use the normal distribution to make a confidence interval for $p_1 - p_2$.

Example 10-13: Solution

b) The standard deviation of $\hat{p}_1 - \hat{p}_2$ is

$$s_{\hat{p}_1 - \hat{p}_2} = \sqrt{\frac{\hat{p}_1 \hat{q}_1}{n_1} + \frac{\hat{p}_2 \hat{q}_2}{n_2}} = \mathbf{.02593742}$$

$$z = 2.17$$

$$\begin{aligned} (\hat{p}_1 - \hat{p}_2) \pm z s_{\hat{p}_1 - \hat{p}_2} &= (.20 - .17) \pm 2.17(.02593742) \\ &= \mathbf{.03 \pm .056 = -.026 \text{ to } .086} \end{aligned}$$

Hypothesis Testing About $p_1 - p_2$

Test Statistic z for $\hat{p}_1 - \hat{p}_2$

The value of the **test statistic z for** $\hat{p}_1 - \hat{p}_2$ is calculated as

$$z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{S_{\hat{p}_1 - \hat{p}_2}}$$

The value of $p_1 - p_2$ is substituted from H_0 , which is usually zero.

Hypothesis Testing About $p_1 - p_2$

$$\sigma_{\hat{p}_1 - \hat{p}_2} = \sqrt{\frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}}$$

$$\bar{p} = \frac{x_1 + x_2}{n_1 + n_2} \quad \text{or} \quad \frac{n_1 \hat{p}_1 + n_2 \hat{p}_2}{n_1 + n_2}$$

$$s_{\hat{p}_1 - \hat{p}_2} = \sqrt{\bar{p} \bar{q} \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

where $\bar{q} = 1 - \bar{p}$

Example 10-14

Reconsider Example 10-13 about the percentages of users of two toothpastes who will never switch to another toothpaste. At the 1% significance level, can we conclude that the proportion of users of Toothpaste A who will never switch to another toothpaste is higher than the proportion of users of Toothpaste B who will never switch to another toothpaste?

Example 10-14: Solution

$$\hat{p}_1 = x_1 / n_1 = 100 / 500 = .20$$

$$\hat{p}_2 = x_2 / n_2 = 68 / 400 = .17$$

Example 10-14: Solution

- Step 1:
- $H_0: p_1 - p_2 = 0$
 - p_1 is not greater than p_2
- $H_1: p_1 - p_2 > 0$
 - p_1 is greater than p_2

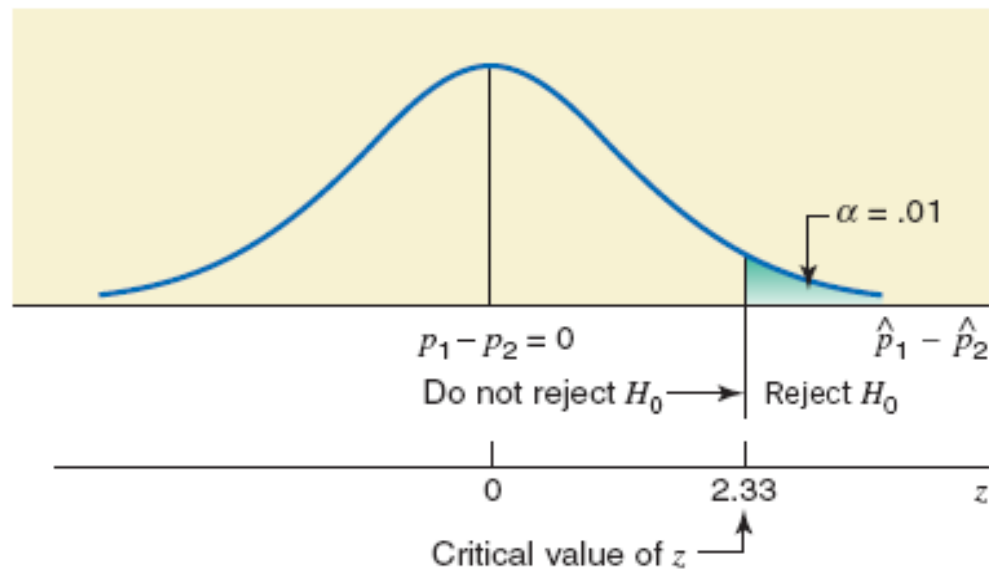
Example 10-14: Solution

- Step 2:
- $n_1\hat{p}_1, n_1\hat{q}_1, n_2\hat{p}_2,$ and $n_2\hat{q}_2$ are all greater than 5
- Both samples sizes are large
- We use the normal distribution to make the test

Example 10-14: Solution

- Step 3:
- The $>$ sign in the alternative hypothesis indicates that the test is right-tailed.
- $\alpha = .01$
- The critical value of z is 2.33

Figure 10.8



Example 10-14: Solution

□ Step 4:

$$\bar{p} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{100 + 68}{500 + 400} = .187$$

$$\bar{q} = 1 - \bar{p} = 1 - .187 = .813$$

$$s_{\hat{p}_1 - \hat{p}_2} = \sqrt{\bar{p}\bar{q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)} = \sqrt{(.187)(.813)\left(\frac{1}{500} + \frac{1}{400}\right)} = .02615606$$

$$z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{s_{\hat{p}_1 - \hat{p}_2}} = \frac{(.20 - .17) - 0}{.02615606} = 1.15$$

Example 10-14: Solution

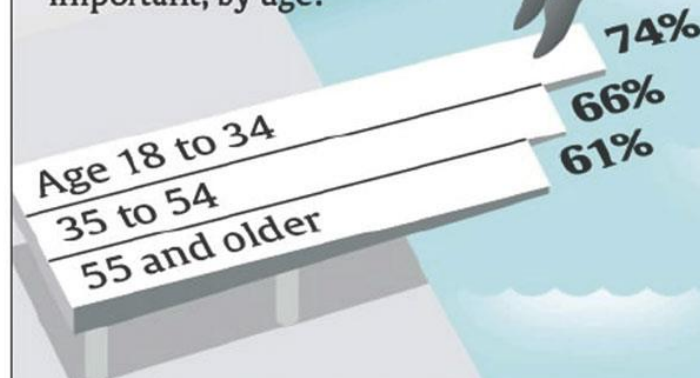
- Step 5:
- The value of the test statistic $z = 1.15$
 - It falls in the nonrejection region
- We fail to reject the null hypothesis
- Therefore, We conclude that the proportion of users of Toothpaste A who will never switch to another toothpaste is not greater than the proportion of users of Toothpaste B who will never switch to another toothpaste.

Case Study 10-2 Is Vacation Important?

USA TODAY Snapshots®

Vacation more important for younger generation

Percentage who said taking a vacation is very important, by age:



Source: Access America and Ipsos Public Affairs survey of 1,000 adults 18 and older. Margin of error ± 3 percentage points

By Jae Yang and Sam Ward, USA TODAY

Example 10-15

According to a July 1, 2009, Quinnipiac University poll, 62% of adults aged 18 to 34 years and 50% of adults aged 35 years and older surveyed believed that it is the government's responsibility to make sure that everyone in the United States has adequate health care. The survey included approximately 683 people in the 18- to 34-year age group and 2380 people aged 35 years and older. Test whether the proportions of people who believe that it is the government's responsibility to make sure that everyone in the United States has adequate health care are different for the two age groups. Use the 1% significance level.

Example 10-15: Solution

- Step 1:
- $H_0: p_1 - p_2 = 0$
 - The two population proportions are not different
- $H_1: p_1 - p_2 \neq 0$
 - The two population proportions are different

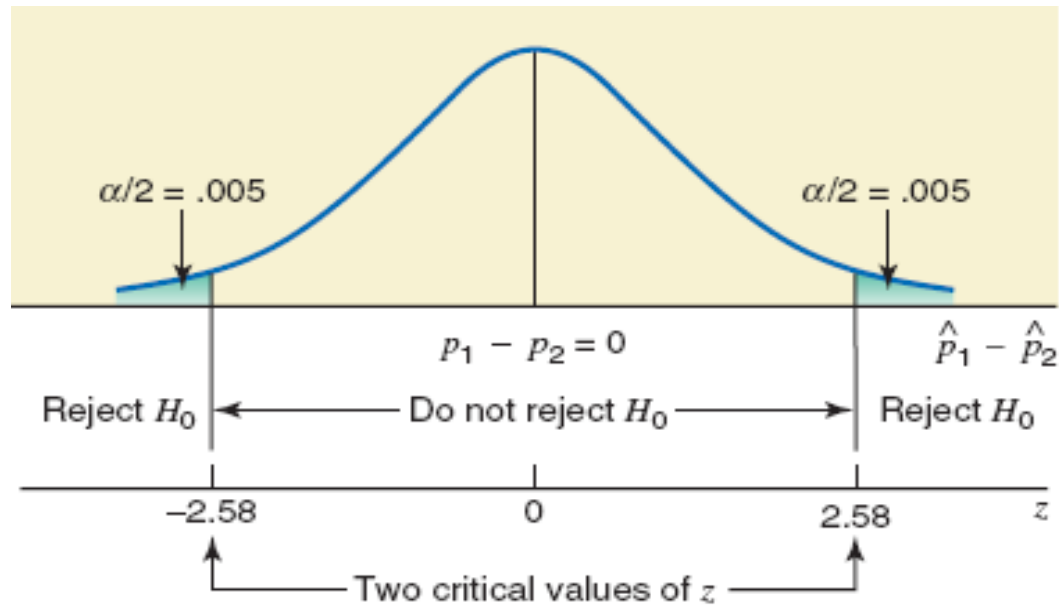
Example 10-15: Solution

- Step 2:
- The samples are large and independent
- We apply the normal distribution
- $n_1\hat{p}_1$, $n_1\hat{q}_1$, $n_2\hat{p}_2$ and $n_2\hat{q}_2$ are all greater than 5 (These should be checked)

Example 10-15: Solution

- Step 3:
- The $\neq$ sign in the alternative hypothesis indicates that the test is two-tailed.
- $\alpha = .01$
- The critical values of z are -2.58 and 2.58

Figure 10.9



Example 10-15: Solution

Step 4:

$$\bar{p} = \frac{n_1 \hat{p}_1 + n_2 \hat{p}_2}{n_1 + n_2} = \frac{683(.62) + 2380(.50)}{683 + 2380} = .527$$

$$\bar{q} = 1 - \bar{p} = 1 - .527 = .473$$

$$s_{\hat{p}_1 - \hat{p}_2} = \sqrt{\bar{p}\bar{q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)} = \sqrt{(.527)(.473)\left(\frac{1}{683} + \frac{1}{2380}\right)} = .02167258$$

$$z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{s_{\hat{p}_1 - \hat{p}_2}} = \frac{(.62 - .50) - 0}{.02167258} = 5.54$$

Example 10-15: Solution

- Step 5:
- The value of the test statistic $z = 5.54$
- It falls in the rejection region
- We reject the null hypothesis H_0
- Therefore, we conclude that the proportions of adults in the two age groups who believe that it is the government's responsibility to make sure that everyone in the US has adequate health care are different.